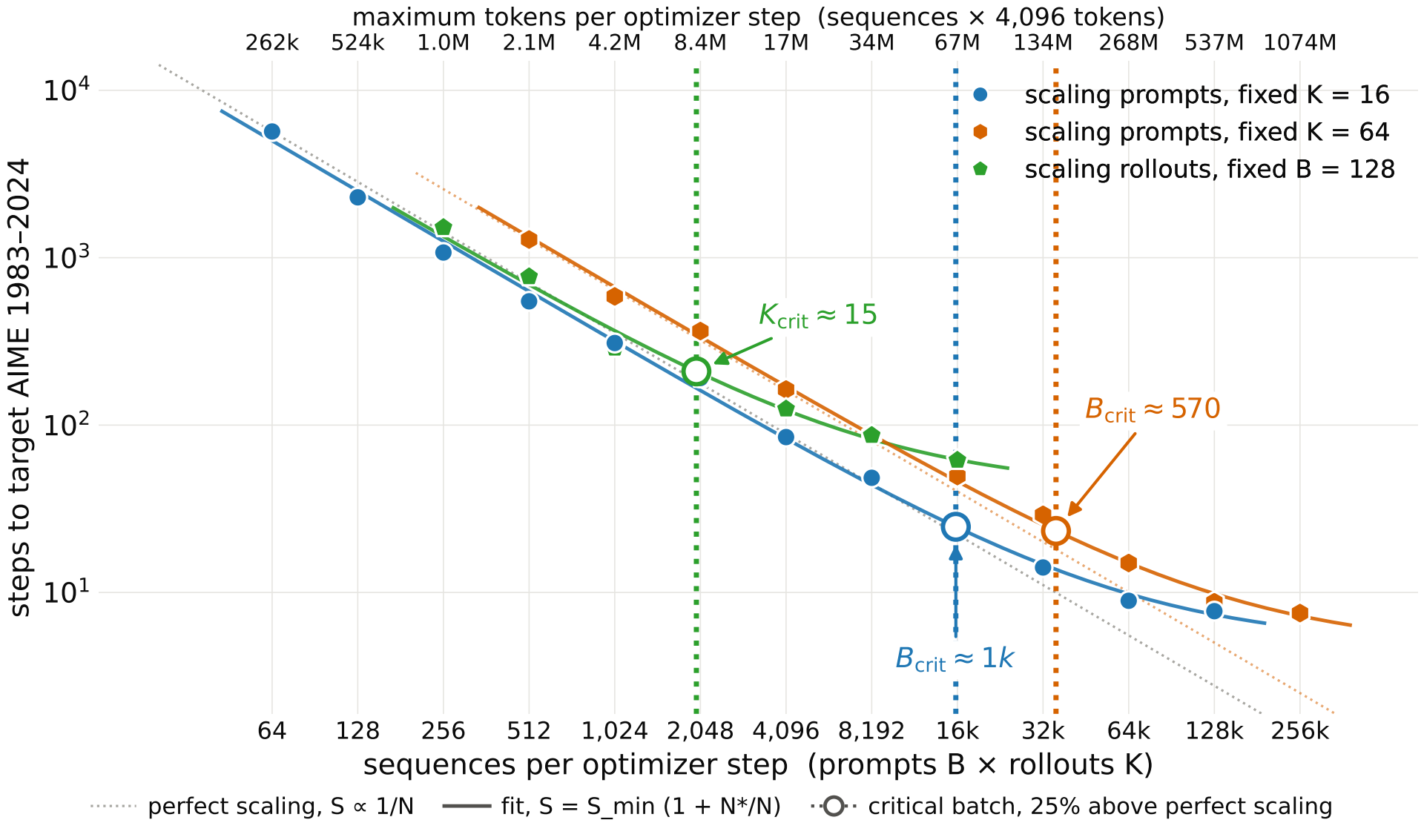
maximum tokens per optimizer step (sequences × 4,096 tokens)
262k
524k
1.0M
2.1M
4.2M
8.4M
17M
34M
67M
134M
268M
537M
1074M
steps to target AIME 1983–2024
10^4
10^3
10^2
10^1
scaling prompts, fixed K = 16
scaling prompts, fixed K = 64
scaling rollouts, fixed B = 128
$K_{crit} \approx 15$
$B_{crit} \approx 570$
$B_{crit} \approx 1k$
64
128
256
512
1,024
2,048
4,096
8,192
16k
32k
64k
128k
256k
sequences per optimizer step (prompts B × rollouts K)
perfect scaling, S ∝ 1/N
fit, S = S_min (1 + N*/N)
critical batch, 25% above perfect scaling